

1. Tocharian

Scoring (max 39)

- **2.1.** 1 point for each correctly ordered stage (assume a stage is correct unless it is located later than a stage which it should precede) (max 7)
- **2.1.** 1 point for each correctly stated change linked to the correct stage. (max 6)
 - Only accept maximally general rules – e.g. ‘j > g’ but not ‘j > g/_o’
 - Only accept maximally simple rules – e.g. ‘b^h > 0/m_’, but not ‘mb^h > m’.
 - Accept plausible alternatives, e.g. ‘s > 0/o_’ or ‘b^h > 0/ not at beginning of word’.
- **2.2.**
 - 2 points for each correct ordering statement; 1 for undergeneralised statements. (max 16)
 - 1 point for each correct justification. (max 8)
 - 2 points for a diagram showing global ordering (max 2)

Q.2.1

order	A-G	Change(s) that turned this stage into the next one
1 (earliest)	D	c > k
2	C	j > g
3	G	b ^h > Ø / m _
4	F	b ^h > p
5	B	g > k
6	A	s > Ø / _ #
7 (latest)	E	

Q.2.2 [Please ask for extra paper if you need it.]

The rules given in the question are repeated below.

1	o > ë	10	m > n / _ t
2	nt > Ø / _ #	11	u > ə
3	e > ə	12	k > ś / _ e, ē
4	or > ur / _ #	13	w > w ^y / _ i, ī
5	ē, ī, ū > e, i, u	14	g > k
6	m > n / _ #	15	d > Ø / _ #
7	t > c / _ e, ē	16	s > Ø / _ #
8	d > ś / _ e, ē	17	ti > Ø / n _ #
9	n > Ø / _ #		

The first thing to realize here is that (as the question suggests) only a partial order can be deduced. For example, 15 can be put anywhere in the ordering, because it doesn’t have an effect on any other rules. The rules that can be ordered are the ones that make reference to each other’s inputs and outputs.

One useful notation for ordering changes is A + B, meaning ‘change A applied before change B’, but other notations are of course possible. It would be tempting, but confusing, to use ‘>’ here as well as within the rule.

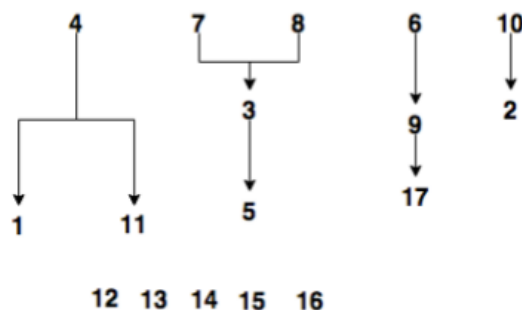
The orderings it is possible to work out (by a kind of modus tollens) are given below.

- 4 applied before both 1 and 11 (4 + 1, 11) [two orderings]

In *akëntər*, we have: or > ur (by 4) > ər (by 11). If the order were 1 + 4 (meaning ‘1 before 4’), we would have: or > ər (by 1) and 4 would fail to apply. If the order were 11 + 4, we would have: or > ur by 4, but then no subsequent change to ər because 11 is no longer available.

- 7 and 8 both applied before 3 (7, 8 + 3) [two orderings]
In *cək^wē*, we have: té > cé (by 7) > cǎ (by 3). If the order were 3 + 7, we would have: té > tá (by 3) and 7 would fail to apply. The same reasoning applies to 8 in the word *śákə* (‘ten’).
- 3 applied before 5 (3 + 5)
In *pacér*, we have: é̇ > é (by 5), but 3 is no longer available to change é to á. If the order were 5 > 3, we would have: é̇ > é (by 5) > á (by 3).
Also *kēruwos* > *ser*...
- 6 applied before 9
In *kántē*, we have: m > n (by 6) > Ø (by 9). If the order were 9 > 6, we would have: -m > -n (by 6), but then no subsequent deletion because 9 is no longer available.
- 10 applied before 2
In *śákə*, we have: əmt > ənt (by 10) > ə (by 2). If the order were 2 > 10, we would have: əmt > ənt (by 10), but then no subsequent deletion of nt because 2 is no longer available.
Or: *dekəmt* > *dekənt* > *səkəmt*
- 9 applied before 17
In *akēn*, we have: nti > n (by 17), where 9 has already applied earlier. If the order were 17 > 9, we would have: nti > n (by 17) > Ø (by 9).
- 12, 13, 14, 15, and 16 are unorderable with respect to any other rules.

A neat diagram of a solution might look like this, where a downward arrow means ‘precedes’.



Comment

Working out an order of the steps relative to one another should be straightforward by any method, e.g. by observing that the first word begins either b^h- or p-, and that the simplest conclusion is that this represents a historical change from one to the other (so all the forms with b^h- will be at ‘one end’ of the timeline).

This will give an ordering of the elements relative to one another (e.g. “this stage goes in between these other two”), but the next step is to tell which part is the beginning and which is the end. The trick to this is to realize that some changes are irreversible, specifically the mergers of c with already existent k (this is why the word *kóro-* is included) and later g with k. The statement in the rubric that sound changes must be regular means that the reverse of this can’t happen; that would require an irregular change of k to g in some words but not in others. (A logical possibility is that k becomes g before on but not or; candidates should realize that this is an implausible analysis.)